

Supplementary Material for Why No Universal Output Metric Can Measure Reasoning Ability

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This supplementary material documents data provenance and analysis coverage, secondary-analysis definitions, complete per-configuration tables, robustness analyses, and architecture-level instantiations used in the main paper. All empirical analyses are derived from the unified prediction records. Several derived matched-separability and residual-feature quantities are reported to three-decimal precision, while the repeated-success analysis retains full precision. Throughout, we treat conventional, controlled, confidence-bounded, and residual-feature-adjusted quantities as observable behavioral criteria rather than as direct measures of reasoning ability or of the unobservable process that produced the outputs.

Additional Key Words and Phrases: reasoning evaluation, metric validity, matched evaluation, task-specific verification, non-semantic information

Organization. Section S1 documents provenance and analysis coverage. Section S2 defines the secondary analyses, including the full-precision group outputs. Sections S3–S5 report aggregate results, sensitivity analyses, and complete constraint tables. Section S6 provides architecture-level instantiations of the main argument, and Section S7 records reproduction notes and evidence boundaries.

S1 Provenance, Analysis Coverage, and Claim Discipline

All empirical analyses are derived from the same unified prediction records for 3-SAT, 2-SAT, and graph coloring. From these records, we compute the matched-separability quantities used for row-level metrics, positive-side verification, confidence bounds, and pair-rematching; the residual-feature sensitivity quantities used for the additional structural-feature analysis; and the full-precision aggregation and repeated-success quantities. A single domain-neutral implementation computes the equal-group, equal-pair, nonlinear, variance, and repeated-success summaries.

The unified records contain 19, 66, and 45 model–problem-size configurations for 3-SAT, 2-SAT, and graph coloring, respectively, for 130 configurations in total. The matched-separability tables reported here contain 19, 59, and 45 configurations, respectively. This difference in analysis coverage does not reflect separate sets of model responses.

The repeated-success analysis also contains one overall row for each of the 15 model configurations. Each group contains at most 11 candidate pairs. The analysis evaluates $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ for the nonlinear summaries and computes $A_{\geq k}$ and $A_{\geq k}^W$ for every $k \in \{1, \dots, 10\}$.

For four 2-SAT models—Llama-Nemotron-Super-49B, QwQ-32B, Qwen3-14B, and Qwen3-32B—the prediction records contain many responses for which a SAT/UNSAT label can be recovered from free text but no final answer object is available from which an assignment can be verified. Among positive predictions for these models, the fraction lacking a verifiable assignment ranges from 24% to 90% in the audit. We therefore mark the affected overall rows with $\dagger$ in the main paper and interpret their W and $A_{\geq k}^W$ values conservatively. This issue does not change the definition of the full 2-SAT operational reference: under $\text{ADR}_{\text{full}}^{\dagger wu}$, a pair lacking required verifiable evidence does not satisfy the criterion.

S2 Secondary-Analysis Definitions

S2.1 Task-Specific Verification Criteria and Two-Sided Separability

For 3-SAT, the operational reference is ADR^{+w} . For graph coloring, it is ADR^{+a} . Both require correct matched labels together with a verified positive-side construction: a satisfying assignment for 3-SAT or a proper coloring for graph coloring. In the matched-separability tables, both quantities appear in the column that this supplement denotes by W .

For 2-SAT, $W = \text{ADR}^{+w}$ is instead the positive-side verification criterion reported in the matched-separability tables. It is a diagnostic criterion, not the 2-SAT operational reference. As in the main paper, the 2-SAT operational reference is $\text{ADR}_{\text{full}}^{+wu}$, computed from the unified prediction records. It requires correct matched labels, a verified SAT-side satisfying assignment, and verified UNSAT-side evidence across all complete pairs.

The 2-SAT data also provide the conditional diagnostic criterion ADR^{+wu} . It requires a verified satisfying assignment for the SAT instance and a solver-verified unsatisfiable clause subset for the UNSAT instance, but it is computed only over pairs for which the additional UNSAT-side response is available and can be verified. By contrast, $\text{ADR}_{\text{full}}^{+wu}$ applies the same two-sided requirement to every complete pair at a given problem size. Pairs with absent, invalid, or unavailable required evidence do not satisfy the full-set criterion. The two quantities therefore have different denominators and should not be interpreted as interchangeable estimates of the same rate.

For each matched group G , let $\hat{\theta}_+(G)$ and $\hat{\theta}_-(G)$ be positive- and negative-class recall. The point two-sided separability is

$$\hat{s}_G = \hat{\theta}_+(G)\hat{\theta}_-(G).$$

The matched-separability analysis reports the group-averaged point estimate together with a factorized Clopper–Pearson lower bound L_{CP} and Wilson lower bound L_W . Because $\hat{s}_G = \text{G-mean}^2$, the point statistic induces the same classifier ordering as G-mean. The contribution of the matched-separability protocol is the matched-group and confidence construction, not a new scalar ordering.

S2.2 Strict Ordering Discordance

Within each task and problem size, consider model pairs for which both a metric M and W differ at the reported precision. We compute

$$D_M = \frac{\sum_N \sum_{a < b} \mathbf{1}[(M_{a,N} - M_{b,N})(W_{a,N} - W_{b,N}) < 0]}{\sum_N \sum_{a < b} \mathbf{1}[M_{a,N} \neq M_{b,N} \wedge W_{a,N} \neq W_{b,N}]}.$$

No hidden precision is used to break ties. The statistic measures agreement with the task-specific criterion W . For 3-SAT and graph coloring, W is the operational reference. For 2-SAT, it is the diagnostic positive-side verification criterion used in the same matched result set. The statistic does not treat W as a direct measure of reasoning ability.

S2.3 Metric-Level Gap

For each row and metric M , we record whether $M > W$, $M = W$, or $M < W$ at the reported precision and report the mean difference $M - W$. The scales have different operational meanings, so the difference is descriptive rather than an estimate of “overstatement” on a common reasoning-ability scale.

S2.4 Pair-Rematching Sensitivity

The matched-separability tables include the original pairing, a maximum-rematching branch, and a minimum-rematching branch. We compute

$$\Delta_{\text{match}} = \text{ADR}_{\text{max}} - \text{ADR}_{\text{min}}.$$

This quantity changes only the matching rule applied to a fixed set of predictions.

S2.5 Confidence and Boundary Sensitivity

We summarize

$$\Delta_{\text{CI}} = |L_{\text{CP}} - L_{\text{W}}|$$

for every constraint row. Threshold-parametric prefix boundaries are read as

$$N^*(\tau) = \max \{N : L_{\text{CP}}(n) \geq \tau \text{ for every tested } n \leq N\},$$

subject to the matched-separability analysis's prefix convention.

The residual-feature sensitivity analysis additionally defines an ideal floor $C_0 = 0.25$ and a measured residual non-semantic ceiling. Its pre-registered residual feature set is

$$\mathcal{F}_{\text{CB}} = \{\text{mean_clause_len}, \text{min_clause_len}, \text{max_clause_len}, \text{pos_ratio}, \text{var_coverage}\}.$$

For each group, the analysis takes the maximum two-sample KS distance over these one-dimensional features, subtracts the 0.95 quantile of a within-group label-permutation reference computed from 200 permutations, and truncates at zero. Calling the resulting estimate $\hat{d}_{\text{short}}$, the ceiling is

$$\widehat{C}_{\text{sc}} = \left(\frac{1 + \hat{d}_{\text{short}}}{2} \right)^2 \geq C_0.$$

The residual-feature analysis uses its own pooled factorized lower bound $\underline{s}_{\text{M}}^{\text{CP}}$:

$$\mathcal{M}_0 = \underline{s}_{\text{M}}^{\text{CP}} - C_0, \quad \mathcal{M}_{\text{sc}} = \underline{s}_{\text{M}}^{\text{CP}} - \widehat{C}_{\text{sc}}.$$

We use these reported boundaries only as a design-sensitivity analysis and do not numerically merge $\underline{s}_{\text{M}}^{\text{CP}}$ with the matched-separability analysis's group-averaged lower bound. These margins do not identify the internal mechanism.

S2.6 Group Aggregation and Repeated-Success Outputs

Let $\mathcal{G}_c$ contain the groups with at least one complete matched pair. For each group g , let n_g be the number of complete pairs, S_g the number whose two labels are both correct, and $a_g = S_g/n_g$. The equal-group and equal-pair summaries are

$$\text{GADR} = \frac{1}{|\mathcal{G}_c|} \sum_{g \in \mathcal{G}_c} a_g, \quad \text{ADR} = \frac{\sum_g S_g}{\sum_g n_g}.$$

For $v_g = a_g(1 - a_g)$ and $\lambda \geq 0$, the analysis also exports

$$\text{SGADR}_\lambda = \frac{1}{|\mathcal{G}_c|} \sum_{g \in \mathcal{G}_c} (a_g - \lambda v_g), \quad \text{SGADR}_{\star, \lambda} = \text{ADR} - \lambda \text{ADR}(1 - \text{ADR}).$$

At $\lambda = 1$, these reduce to the macro-average of a_g^2 and ADR^2 , respectively. We additionally export between-group variance and

$$A_{\geq k} = \frac{1}{|\mathcal{G}_c|} \sum_{g \in \mathcal{G}_c} \mathbf{1}[S_g \geq k], \quad k = 1, \dots, 10.$$

Table 1. Metric level relative to the positive-side verified criterion W . For 3-SAT and graph coloring, W is the operational reference; for 2-SAT, it is the diagnostic positive-side criterion used in this analysis. Counts are computed row by row at the reported precision.

Domain	Metric	Rows	$M > W$	$M = W$	$M < W$	Mean ($M - W$)
3-SAT	Acc	19	18	1	0	0.322
3-SAT	ADR	19	15	4	0	0.077
3-SAT	$\hat{s}$	19	15	2	2	0.079
3-SAT	L_{CP}	19	10	0	9	-0.010
3-SAT	G-mean	19	18	1	0	0.234
3-SAT	BA	19	18	1	0	0.323
3-SAT	MCC	19	14	2	3	0.087
2-SAT	Acc	59	55	4	0	0.319
2-SAT	ADR	59	47	12	0	0.057
2-SAT	$\hat{s}$	59	47	12	0	0.057
2-SAT	L_{CP}	59	32	1	26	0.015
2-SAT	G-mean	59	54	5	0	0.205
2-SAT	BA	59	55	4	0	0.322
2-SAT	MCC	59	43	9	7	0.069
Graph coloring	Acc	45	45	0	0	0.490
Graph coloring	ADR	45	40	5	0	0.155
Graph coloring	$\hat{s}$	45	41	2	2	0.159
Graph coloring	L_{CP}	45	39	2	4	0.114
Graph coloring	G-mean	45	43	2	0	0.359
Graph coloring	BA	45	45	0	0	0.488
Graph coloring	MCC	43	35	2	6	0.134

For the same groups, let S_g^W count pairs that have both labels correct and additionally satisfy the positive-side construction requirement. We define

$$A_{\geq k}^W = \frac{1}{|\mathcal{G}_c|} \sum_{g \in \mathcal{G}_c} \mathbf{1}[S_g^W \geq k], \quad k = 1, \dots, 10.$$

For 3-SAT and graph coloring, this positive-side requirement is the operational reference used in the main paper. For 2-SAT, it is a diagnostic criterion used for the calibration analysis rather than the full two-sided operational reference.

The output bundle keeps the three domains separate and includes overall, model-size, per-group, lambda-sweep, completion, and applicability records. The repeated-success outputs contain 19, 66, and 45 model-size rows for 3-SAT, 2-SAT, and graph coloring, respectively. All of these outputs are computed from the same unified prediction records used by the other empirical analyses.

Table 2. Strict pairwise model-ordering discordance with W . For 3-SAT and graph coloring, W is the operational reference; for 2-SAT, it is the diagnostic positive-side criterion used in this analysis. Each cell is discordant/strict-comparable pairs; ties at the reported precision are excluded.

Domain	Acc	F1 ₊	F1 ₋	ADR	$\hat{s}$	L_{CP}	G-mean	BA	MCC
3-SAT	3/16	4/16	2/16	4/16	4/16	3/16	4/16	3/16	3/16
2-SAT	10/115	25/116	10/111	11/116	12/116	9/116	13/116	12/116	14/116
Graph coloring	25/62	34/62	21/63	24/62	26/63	27/63	30/63	24/62	25/62

Table 3. Sensitivity to Clopper–Pearson versus Wilson lower bounds.

Domain	Mean abs. diff.	Median	95th pct.	Maximum
3-SAT	0.0043	0.0050	0.0073	0.010
2-SAT	0.0050	0.0060	0.0100	0.011
Graph coloring	0.0040	0.0030	0.0100	0.011

Table 4. Pair-rematching sensitivity, $\Delta_{\text{match}} = \text{ADR}_{\text{max}} - \text{ADR}_{\text{min}}$.

Domain	Rows	Mean	Median	Maximum	Nonzero rows
3-SAT	19	0.026	0.021	0.080	78.9%
2-SAT	59	0.023	0.000	0.182	42.4%
Graph coloring	45	0.072	0.019	0.364	73.3%

S3 Aggregate Secondary Results

S3.1 Selected Strict Ordering Reversals

S3.2 Additional Aggregation and Repeated-Success Results

The main paper reports representative overall repeated-success results. The repeated-success analysis also permits model–problem-size sensitivity checks that were omitted from the main text. Table 7 summarizes two of them. Equal-group and equal-pair weighting usually produce similar values, but the maximum difference can still be substantial in an individual configuration. Requiring ten successful pairs is much more selective: $A_{\geq 10} = 0$ in most model–problem-size configurations for all three tasks.

The nonlinear summaries expose another aggregation choice. For Opus 4.7 on 3-SAT, the overall values are $\text{GADR} = 0.600$, $\text{SGADR}_1 = 0.514$, and $\text{SGADR}_{\star,1} = 0.415$. For QwQ-32B on graph coloring, the same sequence is 0.423, 0.285, and 0.179. QwQ-32B on 2-SAT has the largest overall between-group variance, 0.185. These transformations summarize different aspects of heterogeneity in the same prediction records. Changing λ changes the variance penalty rather than the underlying model outputs.

Repeated-success thresholds produce a larger effect. Across the overall rows, $A_{\geq 1}$ reaches 1.000, whereas the largest $A_{\geq 10}$ is 0.340. Qwen3-32B on graph coloring has overall $\text{ADR} = 0.314$ and $A_{\geq 1} = 0.856$, but only $A_{\geq 10} = 0.015$. Opus 4.7 on 3-SAT has $\text{ADR} = 0.644$, while $A_{\geq 10} = 0.224$. Completion is high for most overall rows but ranges from 0.711 to nearly 1, so absolute-count thresholds should be interpreted together with completion. These results support the main paper’s

Table 5. Largest pair-rematching spreads in each domain.

Domain	Model	N	ADR_{pair}	ADR_{max}	ADR_{min}	Δ_{match}
3-SAT	DeepSeek-Reasoner	50	0.090	0.110	0.030	0.080
3-SAT	Opus 4.7 (med. think)	50	0.200	0.220	0.140	0.080
3-SAT	Opus 4.7 (med. think)	60	0.156	0.169	0.117	0.052
3-SAT	DeepSeek-Reasoner	60	0.086	0.086	0.038	0.048
3-SAT	DeepSeek-Reasoner	25	0.289	0.299	0.258	0.041
2-SAT	gpt-oss-20b (reason)	25	0.227	0.364	0.182	0.182
2-SAT	Qwen3-14B (think)	50	0.154	0.231	0.063	0.168
2-SAT	Qwen3-14B (think)	40	0.203	0.238	0.077	0.161
2-SAT	Qwen3-32B (think)	50	0.143	0.185	0.042	0.143
2-SAT	gpt-oss-20b (reason)	10	0.727	0.818	0.682	0.136
Graph coloring	Llama-Nemo-49B (r-on)	60	0.000	0.364	0.000	0.364
Graph coloring	Llama-Nemo-49B (r-on)	50	0.282	0.473	0.191	0.282
Graph coloring	QwQ-32B (think)	40	0.309	0.436	0.155	0.281
Graph coloring	Llama-Nemo-49B (r-on)	40	0.136	0.273	0.000	0.273
Graph coloring	Qwen3-32B (think)	20	0.278	0.389	0.141	0.248

Table 6. Representative model-order reversals relative to W . Arrows indicate the ordering induced by the displayed metric and by W .

Domain/size	Models	Output-metric ordering	W ordering
3-SAT, $N = 25$	Opus 4.7 vs. GPT-5	Acc 0.935 > 0.864; ADR 0.759 > 0.736; $\hat{s}$ 0.783 > 0.732	$W_{\text{Opus}} = 0.655 < W_{\text{GPT}} = 0.673$
Graph coloring, $N = 10$	Llama-Nemo-49B vs. GPT-OSS-20B	Acc 0.909 > 0.886; L_{CP} 0.751 > 0.710	$W_{\text{Llama}} = 0.227 < W_{\text{GPT-OSS}} = 0.818$
Graph coloring, $N = 8$	Llama-Nemo-49B vs. Qwen3-32B	Acc 0.932 > 0.824; ADR 0.864 > 0.647	$W_{\text{Llama}} = 0.136 < W_{\text{Qwen}} = 0.482$
2-SAT, $N = 30$	Qwen3-32B vs. Llama-Nemotron-Super-49B (reason)	ADR 0.157 < 0.250; $\hat{s}$ 0.156 < 0.182	$W_{\text{Qwen}} = 0.083 > W_{\text{Llama}} = 0.000$

interpretation of k as an explicit behavioral requirement rather than a task-independent calibration constant.

S3.3 Positive-Side Verification Under Group Aggregation

Applying a group transformation to label outcomes does not supply the positive-side construction required by W . On the unified prediction records, W is below ADR in every overall row and every model-problem-size configuration. Across the 130 configurations, the mean and median ADR – W gaps are 0.171 and 0.102, respectively. Equal-group weighting has little systematic effect on this

Table 7. Aggregation and repeated-success sensitivity across the unified prediction records. The second and third numeric columns summarize $|GADR - ADR|$. The last column counts configurations with no group containing at least ten label-correct pairs.

Task	Configs.	Mean abs. gap	Max. abs. gap	$A_{\geq 10} = 0$
3-SAT	19	0.0077	0.081	12/19
2-SAT	66	0.0028	0.142	43/66
Graph coloring	45	0.0014	0.017	33/45

Table 8. 3-SAT prefix boundaries $N^*(\tau)$ from the matched-separability analysis.

Model	$N^*(0.50)$	$N^*(0.25)$	$N^*(0.10)$	$N^*(0.05)$
GPT-5	25	30	40	60
Opus 4.7 medium-thinking	25	25	50	60
DeepSeek-Reasoner	10	10	25	25

Table 9. 2-SAT prefix boundaries $N^*(\tau)$ from the matched-separability analysis.

Model	$N^*(0.50)$	$N^*(0.25)$	$N^*(0.10)$	$N^*(0.05)$
Qwen3-14B thinking	15	20	50	50
Qwen3-32B thinking	10	15	20	30
QwQ-32B thinking	10	15	20	25

difference: $GADR - W$ is within 0.03 of $ADR - W$ in 14/15 overall rows. The four marked 2-SAT models are subject to the missing-answer caveat in Section S1.

Numerical proximity produced by a nonlinear penalty should not be interpreted as semantic calibration. $SGADR_1$ and $SGADR_{*,1}$ are numerically closer to W than ADR is in 105/130 and 102/130 configurations, respectively, yet $SGADR_1 < W$ in 40/130 configurations. The quantities differ because $SGADR$ penalizes within-group variation, whereas W removes pairs without a verified positive-side construction. Agreement in value can therefore occur without the two criteria representing the same event.

The same distinction appears in ordering and repeated-success comparisons. Among the six open-weight models, label-only group metrics agree with the ordering induced by W on only 9–10 of the 15 model pairs on graph coloring and 9–13 of 15 on the 2-SAT comparison. For $A_{\geq 10}$, the agreement is 9/15 in both tasks. Moreover, 42 model–problem-size configurations have $A_{\geq 10} > 0$, but 23 of them have $A_{\geq 10}^W = 0$. Sixty-five configurations have $A_{\geq 1} = 1.000$, while their corresponding $A_{\geq 1}^W$ values range down to 0. These results reinforce the main paper’s conclusion that calibration depends on both the group threshold and the pair-level correctness requirement being aggregated. For 2-SAT, these are diagnostic comparisons with the positive-side criterion W , not comparisons with the full operational reference ADR_{full}^{+wu} .

Table 10. Positive-margin boundaries from the residual-feature sensitivity analysis. $\mathcal{M}_0$ uses the ideal 0.25 floor. $\mathcal{M}_{sc}$ uses the measured residual structural-feature ceiling. A dash means no positive boundary on the tested grid.

Task	Model	$\mathcal{M}_0$ boundary	$\mathcal{M}_{sc}$ boundary	Shift
3-SAT	GPT-5	30	25	−5
3-SAT	Opus 4.7 medium-thinking	25	25	0
3-SAT	DeepSeek-Reasoner	10	10	0
2-SAT	Qwen3-32B thinking	15	10	−5
2-SAT	gpt-oss-20b reasoning	20	15	−5
2-SAT	Nemotron-H-47B (r-on)	−	−	−
2-SAT	LN-Super-49B (r-on)	10	8	−2
Graph coloring	Qwen3-32B thinking	10	8	−2
Graph coloring	gpt-oss-20b reasoning	10	10	0
Graph coloring	Nemotron-H-47B (r-on)	15	10	−5
Graph coloring	LN-Super-49B (r-on)	10	10	0

Table 11. Examples in which an ideal-floor point margin is positive but its Clopper–Pearson lower-bound margin is negative.

Task	Model/size	Point $\mathcal{M}_0$	CP $\mathcal{M}_0$
3-SAT	GPT-5, $N = 40$	+0.053	−0.040
2-SAT	gpt-oss-20b, $N = 25$	+0.156	−0.045

S4 Threshold and Residual-Feature Sensitivity

S4.1 Threshold-Parametric Matched-Separability Boundaries

S4.2 Ideal-Floor Versus Residual Non-Semantic Boundaries

S4.3 Point Estimate Versus Confidence Requirement

S5 Complete Constraint Tables

Table 12. Complete 3-SAT model-size table from the matched-separability analysis. $W = \text{ADR}^{+w}$ is the 3-SAT operational reference and requires a verified satisfying assignment in addition to correct matched labels. Values are shown to three-decimal precision.

Model	N	Acc	F1 ₊	F1 ₋	ADR	W	$\hat{s}$	L_{CP}	L_W	BA	MCC	ADR _{pair}	ADR _{max}	ADR _{min}
GPT-5	5	1.000	1.000	1.000	1.000	1.000	1.000	0.929	0.927	1.000	1.000	1.000	1.000	1.000
GPT-5	10	0.990	0.991	0.989	0.989	0.989	0.981	0.897	0.897	0.991	0.980	0.989	0.989	0.989
GPT-5	25	0.864	0.847	0.877	0.736	0.673	0.732	0.612	0.615	0.864	0.745	0.736	0.755	0.727
GPT-5	30	0.735	0.651	0.787	0.486	0.440	0.480	0.367	0.372	0.734	0.533	0.486	0.495	0.468
GPT-5	40	0.645	0.466	0.735	0.309	0.164	0.303	0.210	0.216	0.645	0.393	0.309	0.309	0.291
GPT-5	50	0.573	0.288	0.695	0.173	0.064	0.168	0.099	0.105	0.573	0.242	0.173	0.173	0.145
GPT-5	60	0.555	0.222	0.688	0.127	0.000	0.125	0.067	0.072	0.555	0.210	0.127	0.127	0.118
Opus 4.7 (med. think)	5	0.945	0.947	0.942	0.897	0.813	0.893	0.782	0.784	0.945	0.892	0.897	0.907	0.888
Opus 4.7 (med. think)	10	0.959	0.960	0.958	0.917	0.861	0.919	0.816	0.817	0.959	0.918	0.917	0.926	0.917
Opus 4.7 (med. think)	25	0.935	0.956	0.875	0.759	0.655	0.783	0.592	0.602	0.894	0.838	0.759	0.759	0.759
Opus 4.7 (med. think)	50	0.527	0.620	0.372	0.200	0.060	0.230	0.124	0.131	0.555	0.134	0.200	0.220	0.140
Opus 4.7 (med. think)	60	0.530	0.656	0.256	0.156	0.000	0.152	0.078	0.084	0.549	0.157	0.156	0.169	0.117
DeepSeek-Reasoner	5	0.986	0.986	0.987	0.973	0.973	0.973	0.892	0.892	0.986	0.973	0.973	0.973	0.973
DeepSeek-Reasoner	10	0.876	0.865	0.886	0.780	0.780	0.761	0.641	0.644	0.879	0.774	0.780	0.780	0.770
DeepSeek-Reasoner	25	0.626	0.454	0.716	0.289	0.144	0.290	0.197	0.203	0.633	0.349	0.289	0.299	0.258
DeepSeek-Reasoner	30	0.515	0.154	0.660	0.073	0.010	0.082	0.036	0.041	0.519	0.075	0.073	0.073	0.052
DeepSeek-Reasoner	40	0.524	0.183	0.664	0.093	0.031	0.104	0.048	0.053	0.520	0.072	0.093	0.103	0.082
DeepSeek-Reasoner	50	0.486	0.194	0.622	0.090	0.000	0.102	0.052	0.057	0.496	-0.011	0.090	0.110	0.030
DeepSeek-Reasoner	60	0.509	0.160	0.653	0.086	0.010	0.086	0.039	0.044	0.513	0.049	0.086	0.086	0.038

Table 13. Complete 2-SAT model-size table from the matched-separability analysis. $W = \text{ADR}^{+w}$ is the diagnostic positive-side 2-SAT verification criterion used in this analysis. It is not the full 2-SAT operational reference $\text{ADR}^{+wu}_{\text{full}}$. Values are shown to three-decimal precision.

Model	N	Acc	$F1_+$	$F1_-$	ADR	W	$\hat{s}$	L_{CP}	L_W	BA	MCC	ADR_{pair}	ADR_{max}	ADR_{min}
gpt-oss-20b (reason)	3	1.000	1.000	1.000	1.000	1.000	1.000	0.976	0.969	1.000	1.000	1.000	1.000	1.000
gpt-oss-20b (reason)	5	1.000	1.000	1.000	1.000	1.000	1.000	0.976	0.969	1.000	1.000	1.000	1.000	1.000
gpt-oss-20b (reason)	8	0.955	0.952	0.957	0.909	0.864	0.909	0.863	0.855	0.955	0.913	0.909	0.909	0.909
gpt-oss-20b (reason)	10	0.841	0.844	0.837	0.727	0.682	0.707	0.631	0.620	0.841	0.683	0.727	0.818	0.682
gpt-oss-20b (reason)	15	0.773	0.737	0.800	0.545	0.500	0.562	0.485	0.475	0.773	0.567	0.545	0.636	0.545
gpt-oss-20b (reason)	20	0.773	0.706	0.815	0.545	0.409	0.545	0.467	0.457	0.773	0.612	0.545	0.545	0.545
gpt-oss-20b (reason)	25	0.568	0.457	0.642	0.227	0.182	0.281	0.216	0.210	0.568	0.149	0.227	0.364	0.182
gpt-oss-20b (reason)	30	0.568	0.296	0.689	0.136	0.045	0.169	0.118	0.116	0.568	0.215	0.136	0.182	0.136
gpt-oss-20b (reason)	40	0.545	0.286	0.667	0.182	0.000	0.165	0.112	0.110	0.545	0.132	0.182	0.182	0.091
gpt-oss-20b (reason)	50	0.500	0.083	0.656	0.045	0.000	0.045	0.026	0.026	0.500	0.000	0.045	0.045	0.045
Qwen3-14B (think)	3	0.990	0.990	0.990	0.981	0.981	0.981	0.950	0.943	0.990	0.981	0.981	0.981	0.981
Qwen3-14B (think)	5	0.974	0.973	0.975	0.948	0.942	0.948	0.909	0.901	0.974	0.949	0.948	0.948	0.948
Qwen3-14B (think)	8	0.974	0.973	0.975	0.948	0.942	0.948	0.907	0.899	0.974	0.949	0.948	0.948	0.948
Qwen3-14B (think)	10	0.942	0.938	0.945	0.883	0.877	0.883	0.833	0.824	0.942	0.889	0.883	0.883	0.883
Qwen3-14B (think)	15	0.844	0.815	0.865	0.688	0.656	0.688	0.620	0.611	0.844	0.724	0.688	0.688	0.688
Qwen3-14B (think)	20	0.669	0.519	0.748	0.338	0.286	0.348	0.282	0.276	0.669	0.432	0.338	0.357	0.338
Qwen3-14B (think)	25	0.604	0.371	0.711	0.214	0.156	0.230	0.179	0.175	0.604	0.309	0.214	0.234	0.208
Qwen3-14B (think)	30	0.516	0.280	0.636	0.162	0.058	0.153	0.105	0.103	0.516	0.043	0.162	0.188	0.078
Qwen3-14B (think)	40	0.490	0.338	0.585	0.203	0.000	0.184	0.136	0.133	0.495	-0.012	0.203	0.238	0.077
Qwen3-14B (think)	50	0.479	0.313	0.580	0.154	0.000	0.163	0.118	0.115	0.479	-0.048	0.154	0.231	0.063
Qwen3-32B (think)	3	0.979	0.979	0.980	0.959	0.959	0.959	0.922	0.914	0.979	0.959	0.959	0.959	0.959
Qwen3-32B (think)	5	0.955	0.952	0.957	0.909	0.909	0.909	0.864	0.855	0.955	0.913	0.909	0.909	0.909
Qwen3-32B (think)	8	0.926	0.920	0.931	0.851	0.843	0.851	0.798	0.790	0.926	0.861	0.851	0.851	0.851
Qwen3-32B (think)	10	0.905	0.895	0.913	0.810	0.777	0.810	0.753	0.744	0.905	0.825	0.810	0.810	0.810
Qwen3-32B (think)	15	0.748	0.663	0.799	0.496	0.430	0.496	0.422	0.413	0.748	0.574	0.496	0.496	0.496
Qwen3-32B (think)	20	0.624	0.397	0.727	0.248	0.207	0.248	0.189	0.185	0.624	0.376	0.248	0.248	0.248
Qwen3-32B (think)	25	0.568	0.246	0.698	0.142	0.092	0.140	0.098	0.096	0.567	0.254	0.142	0.142	0.133
Qwen3-32B (think)	30	0.574	0.270	0.700	0.157	0.083	0.156	0.112	0.110	0.574	0.270	0.157	0.157	0.149
Qwen3-32B (think)	40	0.525	0.173	0.667	0.099	0.000	0.096	0.067	0.066	0.525	0.094	0.099	0.099	0.083
Qwen3-32B (think)	50	0.467	0.273	0.579	0.143	0.000	0.129	0.089	0.087	0.464	-0.083	0.143	0.185	0.042
QwQ-32B (think)	3	1.000	1.000	1.000	1.000	1.000	1.000	0.976	0.969	1.000	1.000	1.000	1.000	1.000
QwQ-32B (think)	5	0.990	0.990	0.990	0.980	0.980	0.980	0.949	0.942	0.990	0.980	0.980	0.980	0.980
QwQ-32B (think)	8	0.980	0.980	0.980	0.960	0.960	0.960	0.924	0.916	0.980	0.960	0.960	0.960	0.960
QwQ-32B (think)	10	0.949	0.947	0.952	0.899	0.869	0.899	0.857	0.849	0.949	0.904	0.899	0.899	0.899
QwQ-32B (think)	15	0.798	0.747	0.832	0.596	0.545	0.596	0.521	0.511	0.798	0.652	0.596	0.596	0.596
QwQ-32B (think)	20	0.641	0.441	0.736	0.283	0.242	0.283	0.222	0.218	0.641	0.406	0.283	0.283	0.283
QwQ-32B (think)	25	0.576	0.276	0.700	0.162	0.111	0.161	0.122	0.119	0.576	0.270	0.162	0.162	0.152
QwQ-32B (think)	30	0.515	0.059	0.673	0.030	0.000	0.030	0.017	0.017	0.515	0.124	0.030	0.030	0.030
QwQ-32B (think)	40	0.487	0.093	0.642	0.033	0.011	0.039	0.018	0.019	0.509	0.043	0.033	0.044	0.022

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Table 13 continued

Model	N	Acc	F1 ₊	F1 ₋	ADR	W	$\hat{s}$	L_{CP}	L_W	BA	MCC	ADR _{pair}	ADR _{max}	ADR _{min}
QwQ-32B (think)	50	0.506	0.044	0.667	0.023	0.000	0.023	0.013	0.013	0.506	0.044	0.023	0.023	0.023
Llama-Nemotron-Super-49B (reason)	3	1.000	1.000	1.000	1.000	0.909	1.000	0.976	0.969	1.000	1.000	1.000	1.000	1.000
Llama-Nemotron-Super-49B (reason)	5	1.000	1.000	1.000	1.000	1.000	1.000	0.976	0.969	1.000	1.000	1.000	1.000	1.000
Llama-Nemotron-Super-49B (reason)	8	0.864	0.842	0.880	0.727	0.727	0.727	0.653	0.642	0.864	0.756	0.727	0.727	0.727
Llama-Nemotron-Super-49B (reason)	10	0.773	0.706	0.815	0.545	0.545	0.545	0.467	0.457	0.773	0.612	0.545	0.545	0.545
Llama-Nemotron-Super-49B (reason)	15	0.727	0.625	0.786	0.455	0.364	0.455	0.377	0.369	0.727	0.542	0.455	0.455	0.455
Llama-Nemotron-Super-49B (reason)	20	0.818	0.778	0.846	0.636	0.545	0.636	0.558	0.548	0.818	0.683	0.636	0.636	0.636
Llama-Nemotron-Super-49B (reason)	25	0.773	0.706	0.815	0.545	0.182	0.545	0.467	0.457	0.773	0.612	0.545	0.545	0.545
Llama-Nemotron-Super-49B (reason)	30	0.400	0.308	0.471	0.250	0.000	0.182	0.094	0.095	0.591	0.237	0.250	0.250	0.250
Nemotron-H-47B-Reasoning (reason)	3	0.524	0.657	0.224	0.077	0.046	0.080	0.069	0.068	0.524	0.076	0.077	0.087	0.074
Nemotron-H-47B-Reasoning (reason)	5	0.510	0.668	0.066	0.023	0.004	0.027	0.019	0.019	0.510	0.062	0.023	0.031	0.021
Nemotron-H-47B-Reasoning (reason)	8	0.499	0.666	–	0.000	0.000	0.000	0.000	0.000	0.499	-0.031	0.000	0.000	0.000
Nemotron-H-47B-Reasoning (reason)	10	0.497	0.664	0.004	0.002	0.000	0.002	0.001	0.001	0.497	-0.042	0.002	0.002	0.002
Nemotron-H-47B-Reasoning (reason)	15	0.513	0.668	0.087	0.027	0.000	0.030	0.024	0.024	0.513	0.070	0.027	0.031	0.025
Nemotron-H-47B-Reasoning (reason)	20	0.510	0.664	0.093	0.043	0.000	0.043	0.037	0.036	0.510	0.049	0.043	0.046	0.041
Nemotron-H-47B-Reasoning (reason)	25	0.526	0.666	0.183	0.070	0.000	0.071	0.058	0.057	0.526	0.096	0.070	0.079	0.062
Nemotron-H-47B-Reasoning (reason)	30	0.503	0.665	0.034	0.015	0.000	0.017	0.012	0.012	0.503	0.024	0.015	0.017	0.015
Nemotron-H-47B-Reasoning (reason)	40	0.527	0.673	0.147	0.074	0.000	0.074	0.055	0.054	0.527	0.120	0.074	0.081	0.062
Nemotron-H-47B-Reasoning (reason)	50	0.516	0.667	0.120	0.046	0.000	0.052	0.042	0.041	0.516	0.076	0.046	0.062	0.041
Nemotron-H-47B-Reasoning (reason)	60	0.537	0.674	0.206	0.105	0.000	0.107	0.085	0.083	0.535	0.125	0.105	0.119	0.093

Table 14. Complete graph-coloring model–size table from the matched-separability analysis. $W = \text{ADR}^{+a}$ is the graph-coloring operational reference and requires a verified proper coloring in addition to correct matched labels. Values are shown to three-decimal precision.

Model	N	Acc	$F1_+$	$F1_-$	ADR	W	$\hat{s}$	L_{CP}	L_W	BA	MCC	ADR_{pair}	ADR_{max}	ADR_{min}
GPT-OSS-20B (think)	8	0.953	0.952	0.955	0.905	0.857	0.909	0.852	0.843	0.955	0.911	0.905	0.905	0.905
GPT-OSS-20B (think)	10	0.886	0.878	0.894	0.818	0.818	0.781	0.710	0.700	0.886	0.780	0.818	0.818	0.773
GPT-OSS-20B (think)	15	0.545	0.286	0.667	0.182	0.136	0.165	0.130	0.127	0.545	0.132	0.182	0.182	0.136
GPT-OSS-20B (think)	20	0.659	0.462	0.750	0.316	0.316	0.313	0.242	0.236	0.650	0.424	0.322	0.322	0.322
GPT-OSS-20B (think)	30	0.528	–	0.691	0.000	0.000	0.000	0.000	0.000	0.500	–	0.000	0.000	0.000
GPT-OSS-20B (think)	40	0.571	0.118	0.717	0.071	0.000	0.062	0.036	0.036	0.531	0.187	0.062	0.062	0.062
GPT-OSS-20B (think)	50	0.440	0.222	0.562	0.182	0.000	0.149	0.098	0.096	0.481	-0.053	0.182	0.182	0.000
GPT-OSS-20B (think)	60	0.550	0.308	0.667	0.111	0.000	0.180	0.120	0.117	0.550	0.140	0.111	0.222	0.111
Llama-Nemo-49B (r-on)	8	0.932	0.927	0.936	0.864	0.136	0.864	0.802	0.792	0.932	0.872	0.864	0.864	0.864
Llama-Nemo-49B (r-on)	10	0.909	0.900	0.917	0.818	0.227	0.818	0.751	0.740	0.909	0.832	0.818	0.818	0.818
Llama-Nemo-49B (r-on)	15	0.545	0.231	0.677	0.136	0.091	0.124	0.092	0.090	0.545	0.158	0.136	0.136	0.091
Llama-Nemo-49B (r-on)	20	0.545	0.167	0.688	0.091	0.045	0.091	0.063	0.062	0.545	0.218	0.091	0.091	0.091
Llama-Nemo-49B (r-on)	30	0.545	0.444	0.615	0.182	0.000	0.264	0.209	0.205	0.545	0.098	0.182	0.364	0.182
Llama-Nemo-49B (r-on)	40	0.364	0.364	0.364	0.136	0.000	0.132	0.087	0.085	0.364	-0.273	0.136	0.273	0.000
Llama-Nemo-49B (r-on)	50	0.632	0.682	0.562	0.312	0.000	0.337	0.249	0.243	0.622	0.244	0.282	0.473	0.191
Llama-Nemo-49B (r-on)	60	0.409	0.381	0.435	0.000	0.000	0.165	0.112	0.110	0.409	-0.183	0.000	0.364	0.000
Nemo-H-47B (r-on)	8	0.668	0.594	0.720	0.338	0.002	0.346	0.293	0.288	0.668	0.362	0.338	0.356	0.337
Nemo-H-47B (r-on)	10	0.641	0.638	0.645	0.301	0.001	0.302	0.251	0.246	0.641	0.283	0.301	0.321	0.286
Nemo-H-47B (r-on)	15	0.582	0.482	0.650	0.193	0.000	0.202	0.160	0.157	0.583	0.179	0.193	0.227	0.171
Nemo-H-47B (r-on)	20	0.556	0.406	0.646	0.144	0.000	0.153	0.121	0.118	0.556	0.131	0.144	0.173	0.130
Nemo-H-47B (r-on)	30	0.537	0.346	0.641	0.116	0.000	0.126	0.097	0.095	0.537	0.090	0.116	0.152	0.093
Nemo-H-47B (r-on)	40	0.505	0.067	0.663	0.021	0.000	0.022	0.017	0.016	0.505	0.029	0.021	0.028	0.017
Nemo-H-47B (r-on)	50	0.503	0.030	0.666	0.009	0.000	0.009	0.007	0.007	0.503	0.027	0.009	0.010	0.008
Nemo-H-47B (r-on)	60	0.500	–	0.667	0.000	0.000	0.000	0.000	0.000	0.500	–	0.000	0.000	0.000
Qwen3-14B (think)	8	0.853	0.829	0.871	0.706	0.473	0.708	0.638	0.629	0.853	0.737	0.706	0.711	0.706
Qwen3-14B (think)	10	0.755	0.680	0.802	0.517	0.351	0.514	0.445	0.436	0.757	0.585	0.515	0.518	0.515
Qwen3-14B (think)	15	0.568	0.478	0.632	0.311	0.096	0.297	0.233	0.227	0.570	0.150	0.311	0.385	0.177
Qwen3-14B (think)	20	0.516	0.604	0.379	0.225	0.000	0.212	0.157	0.153	0.516	0.037	0.225	0.290	0.093
Qwen3-14B (think)	30	0.501	0.660	0.066	0.030	0.000	0.034	0.020	0.020	0.501	0.007	0.030	0.035	0.028
Qwen3-14B (think)	40	0.500	0.657	0.079	0.035	0.000	0.041	0.023	0.023	0.500	0.000	0.035	0.043	0.023
Qwen3-14B (think)	50	0.539	0.693	0.072	0.039	0.000	0.038	0.021	0.021	0.503	0.015	0.039	0.039	0.026
Qwen3-32B (think)	8	0.824	0.788	0.850	0.647	0.482	0.648	0.576	0.566	0.823	0.687	0.645	0.655	0.645
Qwen3-32B (think)	10	0.692	0.564	0.762	0.389	0.308	0.393	0.325	0.318	0.692	0.474	0.389	0.399	0.384
Qwen3-32B (think)	15	0.580	0.370	0.685	0.225	0.096	0.224	0.175	0.171	0.581	0.220	0.223	0.245	0.173
Qwen3-32B (think)	20	0.548	0.475	0.604	0.284	0.046	0.272	0.210	0.205	0.550	0.104	0.278	0.389	0.141
Qwen3-32B (think)	30	0.497	0.586	0.361	0.214	0.000	0.199	0.144	0.141	0.497	-0.006	0.214	0.283	0.059
Qwen3-32B (think)	40	0.505	0.667	0.041	0.021	0.000	0.021	0.012	0.012	0.505	0.043	0.021	0.021	0.016
Qwen3-32B (think)	50	0.583	0.722	0.167	0.091	0.000	0.091	0.052	0.052	0.545	0.227	0.091	0.091	0.091
QwQ-32B (think)	8	0.919	0.912	0.925	0.837	0.407	0.837	0.784	0.775	0.919	0.849	0.837	0.837	0.837
QwQ-32B (think)	10	0.831	0.796	0.855	0.661	0.331	0.661	0.588	0.578	0.831	0.703	0.661	0.661	0.661

Continued on next page

Table 14 continued														
Model	N	Acc	F1 ₊	F1 ₋	ADR	W	$\hat{s}$	L_{CP}	L_W	BA	MCC	ADR _{pair}	ADR _{max}	ADR _{min}
QwQ-32B (think)	15	0.598	0.339	0.711	0.206	0.115	0.204	0.155	0.151	0.598	0.316	0.206	0.206	0.196
QwQ-32B (think)	20	0.566	0.234	0.697	0.132	0.041	0.132	0.089	0.087	0.566	0.266	0.132	0.132	0.132
QwQ-32B (think)	30	0.512	0.224	0.645	0.124	0.008	0.113	0.080	0.078	0.512	0.037	0.124	0.140	0.050
QwQ-32B (think)	40	0.559	0.554	0.565	0.309	0.000	0.298	0.233	0.228	0.560	0.121	0.309	0.436	0.155
QwQ-32B (think)	50	0.500	0.645	0.154	0.091	0.000	0.083	0.046	0.046	0.500	0.000	0.091	0.091	0.000

S6 Architecture-Level Instantiations and Intervention Diagnostics

The main results of this paper are architecture independent. They apply whenever model behavior is evaluated through binary predictions or scalar scores. This section illustrates how the same limitations arise in three familiar model families: additive classifiers, attention-based models, and graph neural networks (GNNs). These examples do not establish additional impossibility results. Their purpose is to show why architectural complexity does not make a higher prediction metric value a reliable indicator of stronger reasoning.

Across these model families, semantic and non-semantic information can both affect the final score. If non-semantic information is more predictive on the benchmark distribution, a model that exploits it can outperform a model that attempts to capture the intended semantic relationship but does so imperfectly. The resulting metric ordering is valid as an ordering of benchmark performance, but it need not be an ordering of reasoning ability.

S6.1 Additive Classifiers over Feature Blocks

We first consider an additive classifier because the contributions of semantic and non-semantic information can be represented explicitly. This is a grouped abstraction of familiar linear, logistic, and additive models [1–3, 9].

Following the semantic/non-semantic decomposition in the main paper, let z denote the semantic factor that should determine the label, and let c denote non-semantic information that may also be available to the model. Let

$$a_z = \phi_z(z), \quad a_c = \phi_c(c)$$

be feature vectors derived from these components. Consider a model f with score

$$s_f(x) = w_z^\top a_z + w_c^\top a_c + b$$

and binary prediction

$$\hat{y} = 1[s_f(x) \geq 0].$$

Define the two score contributions by

$$S_z(x) = w_z^\top a_z, \quad S_c(x) = w_c^\top a_c.$$

Then

$$s_f(x) = S_z(x) + S_c(x) + b.$$

This representation makes two earlier results concrete. First, distinct semantic and non-semantic contributions can yield the same scalar score and therefore the same prediction and ranking. This illustrates the mechanism non-identifiability discussed in the main paper. Second, a larger non-semantic contribution may improve benchmark predictions or rankings even when it does not improve semantic reasoning.

For example, consider a semantic classifier and a non-semantic classifier with class-conditional error rates

$$(\delta_1, \delta_2) \quad \text{and} \quad (\epsilon_1, \epsilon_2),$$

respectively. As shown in Proposition 1 of the main paper, the non-semantic classifier obtains higher accuracy whenever

$$\pi\epsilon_1 + (1 - \pi)\epsilon_2 < \pi\delta_1 + (1 - \pi)\delta_2,$$

where $\pi = \Pr[Y = 1]$. Analogous comparisons hold for balanced accuracy, G-mean, Youden's J , F1, and other prediction metrics.

The additive model makes the interpretation especially clear. The non-semantic classifier is ranked higher because its features produce a more favorable benchmark error profile. The higher

metric value does not imply that the classifier has captured the semantic relationship that determines the labels.

A similar conclusion applies to score-based metrics. If the non-semantic contribution ranks positive and negative benchmark instances more reliably than the semantic contribution, it can increase AUROC. Thus, observing the unthresholded score avoids one source of information loss, but it does not make metric ordering monotone with reasoning ability.

S6.2 Attention-Based Models

Attention-based models provide a less explicit version of the same phenomenon. Let an input be

$$x = (t_1, \dots, t_n),$$

where t_j is the token occurrence at position j . A Transformer maps the input to hidden representations through repeated attention and feed-forward transformations [7]. Abstractly,

$$H^{(0)} = \text{Embed}(x),$$

$$H^{(\ell+1)} = \text{Block}_\ell(H^{(\ell)}), \quad \ell = 0, \dots, L-1.$$

After the final layer, the model constructs an instance representation

$$h_x = \text{Pool}(H^{(L)})$$

and a scalar score

$$s_f(x) = w_{\text{cls}}^\top h_x + b_{\text{cls}}.$$

In software-engineering tasks, some input information bears directly on the intended semantics, while other information may correlate with the labels without determining them. In vulnerability detection, for example, relevant semantic information may include relationships among sources, checks, sanitizers, and sinks. Non-semantic information may include identifier names, comments, API names, formatting conventions, or dataset-construction artifacts. In SAT solving, semantic information is distributed across variables, literals, clauses, and connectives, while generator-specific formatting or structural regularities may provide non-semantic predictive features.

Attention-based models can use either kind of information. A model that captures the relevant semantic relationships may still make errors because those relationships are difficult to learn. Another model may exploit easier non-semantic correlations and obtain a better benchmark error profile or ranking. In that case,

$$M(B, f_c) > M(B, f_z)$$

may hold even though f_z relies more strongly on the intended semantic relationship.

This is the practically important implication of the attention instantiation. A higher accuracy, F1, ADR, or AUROC does not establish that the higher-scoring attention model reasons more effectively. It establishes only that the model performs better according to the observable criterion of the metric on the evaluated distribution.

Attention weights, gradients, and attribution methods such as Integrated Gradients, SHAP, and DeepLIFT can provide additional diagnostic information [4–6]. However, these analyses should not be treated as direct measures of reasoning. Their results depend on the explanation method, the selected representation, and how interactions among tokens are allocated. They can motivate further tests, but they do not by themselves justify interpreting a higher prediction metric as stronger reasoning.

S6.3 Graph Neural Networks and Message Passing

Graph neural networks instantiate the same limitation in models that operate over structured program representations. Let

$$G = (V, E, X)$$

be a program graph, where V is the node set, E is the edge set, and X_v is the initial feature vector of node v . A generic message-passing layer computes

$$m_v^{(k)} = \text{Agg}_{u \in N(v)} \text{Msg}_k(h_v^{(k)}, h_u^{(k)}, e_{uv}),$$

followed by

$$h_v^{(k+1)} = \text{Upd}_k(h_v^{(k)}, m_v^{(k)}).$$

After K layers, the model produces a graph representation

$$h_G = \text{Readout}_{v \in V} h_v^{(K)}$$

and score

$$s_f(G) = w^\top h_G + b.$$

Program graphs may encode AST, control-flow, data-flow, call, or dependence relationships. The semantic condition underlying a vulnerability may depend on a long-range relation, such as an untrusted value reaching a sink without sanitization, a missing bounds check before a memory access, or a particular alias being dereferenced under an unsafe path. These relationships can be difficult to represent and learn.

At the same time, graph inputs may contain easier non-semantic correlations, including risky API names, identifier features, project-specific graph patterns, or local motifs introduced by data extraction. A GNN that exploits these correlations may obtain a higher metric value than a detector that attempts to capture the relevant source-to-sink or control-flow relationship but does so imperfectly.

The resulting ranking is again non-monotone with respect to reasoning ability. A higher metric value can indicate that local or project-specific features are more predictive on the benchmark. It does not necessarily indicate that the model has captured the vulnerability condition represented by the graph.

Graph explanation methods can help diagnose this possibility. For example, GNNExplainer identifies a subgraph and feature subset that approximately preserve a model's prediction [8]. An explanation centered on a vulnerability-relevant slice may provide encouraging evidence, whereas one centered on unrelated names or local graph motifs may indicate potential reliance on non-semantic information. Nevertheless, an explanation describes what influenced the model's output under the assumptions of the explanation method. It does not establish that the model's metric advantage reflects stronger vulnerability reasoning.

S6.4 Intervention-Based Comparison

The model-level examples suggest a common way to strengthen reasoning-oriented evaluation. Rather than interpreting an aggregate metric value in isolation, an evaluator can examine how model outputs respond to controlled semantic and non-semantic changes.

Let $x_{\Delta z}$ be an instance in which the semantic condition is changed while non-semantic information is held as fixed as possible. Let $x_{\Delta c}$ preserve the semantic condition while changing non-semantic information. Define the score changes

$$\Delta_z(x) = |s_f(x) - s_f(x_{\Delta z})|$$

and

$$\Delta_c(x) = |s_f(x) - s_f(x_{\Delta c})|.$$

A model whose outputs track the intended task semantics should generally respond to semantic changes that alter the correct answer and remain comparatively stable under semantics-preserving non-semantic changes. For binary predictions, analogous quantities can be defined through prediction flips:

$$F_z(x) = 1[f(x) \neq f(x_{\Delta z})],$$

$$F_c(x) = 1[f(x) \neq f(x_{\Delta c})].$$

Matched SAT/UNSAT formulas and vulnerable/fixed program variants approximate semantic interventions. Identifier renaming, formatting changes, irrelevant code transformations, and generator perturbations approximate semantics-preserving non-semantic interventions. Pair, group, and cluster metrics summarize success over such controlled instances.

These interventions provide a more informative evaluation design, but the limitations of controlled metrics established in the main paper still apply. A higher controlled score indicates stronger performance under the selected interventions and aggregation rule. It does not automatically imply stronger reasoning, because residual non-semantic correlations, intervention quality, dependence among instances, and threshold selection can still affect the ordering.

S6.5 Summary

The three instantiations differ in how information is represented and combined, but they support the same interpretation. Additive classifiers may combine explicit feature-block contributions. Attention-based models mix information across token representations. GNNs aggregate node and edge information through message passing. These architectural differences can help explain why a model succeeds, but they do not change what a prediction-output metric establishes.

Across all three settings, a higher metric value establishes better performance under the metric's observable benchmark criterion. It does not, without controlled interventions or additional task-specific evidence, establish stronger reasoning. Architecture-specific explanations and attributions can provide useful diagnostics, while verified task-specific constructions and controlled semantic and non-semantic interventions provide stronger behavioral evidence. None of these observations makes an ordinary prediction metric a standalone measure of reasoning ability.

S7 Reproduction Notes and Evidence Boundaries

The accompanying analysis package includes the unified prediction records and CSV outputs for the metric-level, ordering, pair-rematching, confidence-bound, and repeated-success analyses. The matched-separability ordering analysis uses the three-decimal values reported in its derived tables, and ties at that precision are excluded. The repeated-success analysis retains full precision.

The matched-separability protocol describes a matched-pair cluster bootstrap for dependence within seed families. Bootstrap replicates are not reported in the derived analysis outputs used here, so we do not report bootstrap values. The Clopper–Pearson and Wilson bounds reported here are therefore sensitivity summaries rather than cluster-bootstrap estimates.

The operational references are limited by the efficiently verifiable evidence available in each task. For 3-SAT and graph coloring, the study verifies a positive-side construction while the negative side remains label based. For 2-SAT, the full operational reference additionally requires verified UNSAT-side evidence across all complete pairs. These criteria establish what the model successfully demonstrated under the evaluation protocol, not the unobservable reasoning process that produced the outputs.

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